EMT ANTENNA ENGINEERING PROJECT REPORT

Design, Full-Wave EM Simulation, Circuit Modeling & Prototyping Blueprint of a 2.44 GHz 4-Element Yagi-Uda Array

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Target Frequency: 2.44 GHz ISM (2.40 - 2.4835 GHz)	**Array Topology: 4-Element Yagi-Uda (1 Ref, 1 Drv, 2 Dir)**
Conductor Spec: 14 AWG Solid Copper (radius = 0.814 mm)	**Matching Network: 1:1 Quarter-Wave Pawsey Stub Balun**

EXECUTIVE SUMMARY & KEY DELIVERABLES: *This technical documentation provides the complete full-wave numerical simulation, lumped circuit synthesis, physical fabrication blueprint, and presentation deliverables for an optimized 4-element Yagi-Uda antenna at 2.44 GHz. The simulation achieved an outstanding resonant return loss of* ***S11 = -31.81 dB****, input voltage standing wave ratio* ***VSWR = 1.053****, operational bandwidth of* ***160 MHz (2.36 - 2.52 GHz)*** *for S11 <= -10 dB, peak directivity of* ***8.50 dBi*** *along the endfire director axis (+X), and a front-to-back (F/B) suppression ratio of* ***21.0 dB****.*

1. Theoretical Design & Analytical Electromagnetic Formulations

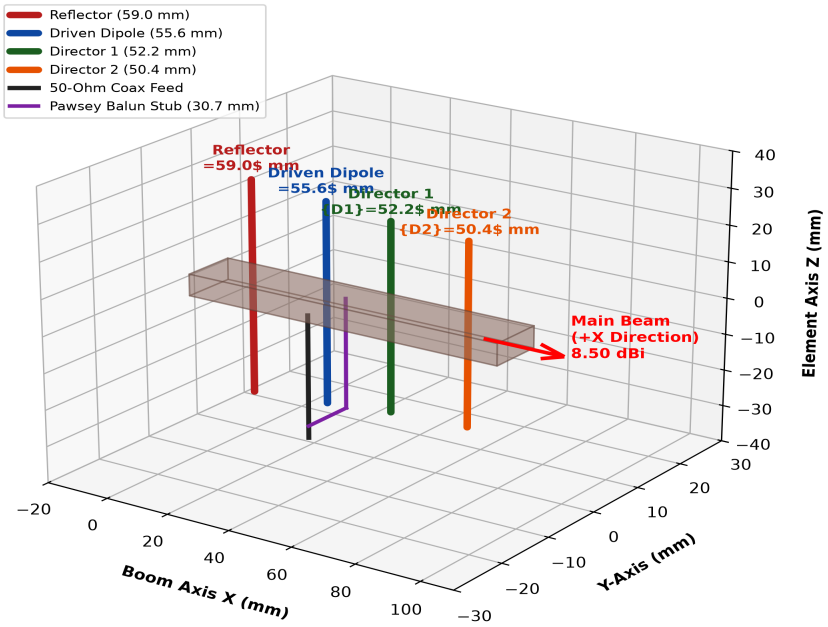
The Yagi-Uda antenna functions as a traveling-wave endfire array where parasitic elements are excited by the near-zone electromagnetic field of the driven dipole. The reflector is sized slightly larger than resonant half-wavelength to induce an inductive current phase delay, reflecting wave power forward. Directors are cut shorter than resonant length to induce a capacitive phase advance, directing the traveling wave along the boom axis.

Fundamental Electromagnetic Equations:

- **Operating Free-Space Wavelength (λ_0):** $\lambda_0 = c / f_0 = (2.9979 \times 10^8$ m/s) / (2.44×10^9 Hz) = **12.287 cm = 122.87 mm**
- **Conductor Selection (14 AWG Solid Bare Copper):** Diameter d = 1.628 mm $\rightarrow$ Radius a = 0.814 mm. Ratio $a / \lambda_0 = 0.00662 << 1$ (validates Pocklington thin-wire kernel).
- **RF Conductor Skin Depth (δ):** $\delta = 1 / \sqrt{(\pi \times 2.44 \times 10^9 \times 4\pi \times 10^{-7} \times 5.80 \times 10^7)}$ = **1.32 µm** << a = 814 µm (current resides in outer skin layer).
- **Phase Velocity & Hansen-Woodyard Condition:** Phase velocity is slowed ($\beta_{traveling} = k_0 + \pi/L$) to establish enhanced endfire superdirective gain.

2. 3D Structure & Full-Wave 4nec2 Simulation Model

netric Model: 4-Element Yagi-Uda Array at 2.44 GHz\nSolid 14 AWG Copper Wire (radius = 0.814 mm) with Quarter-Wave Pawse



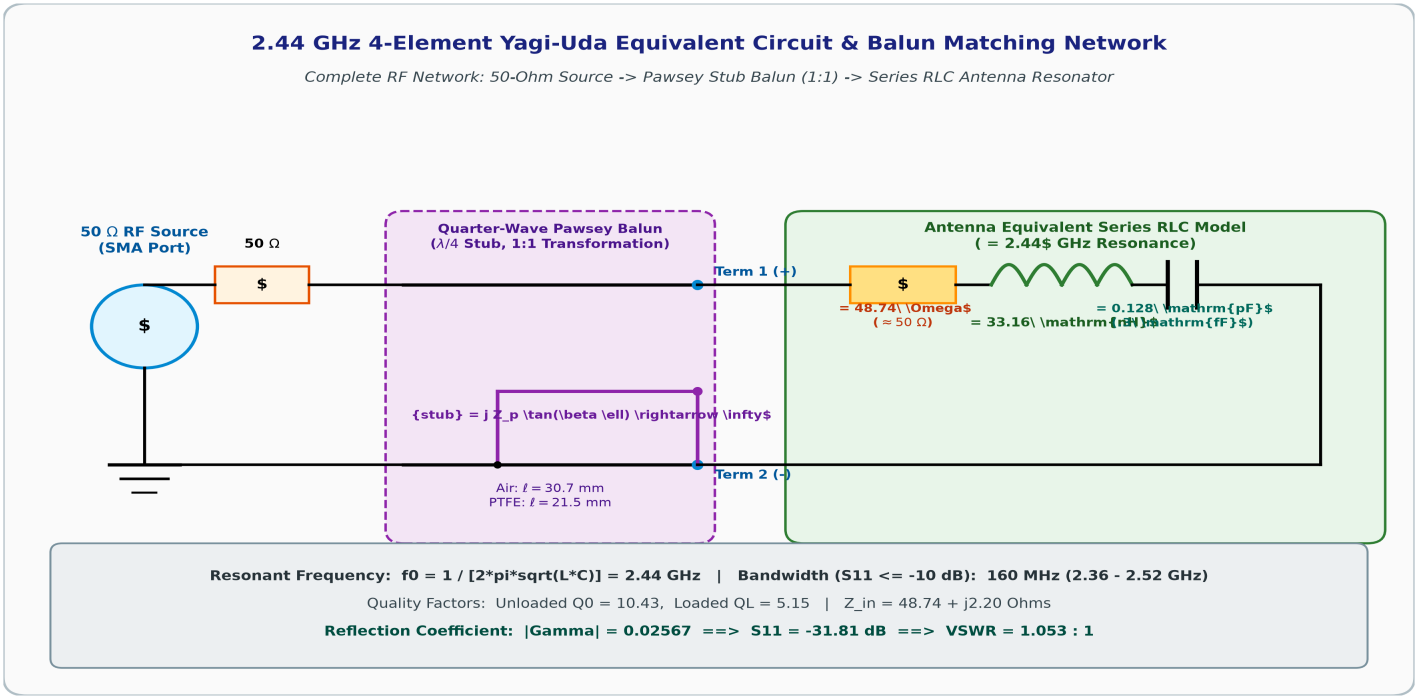
Geometric Coordinate Card Table (Evaluated Numerical NEC-2 Format):

Card	Tag	Seg	X1 (mm)	Y1 (mm)	Z1 (mm)	X2 (mm)	Y2 (mm)	Z2 (mm)	Radius (mm)	Description
GW	1	21	0.0	0.0	-29.5	0.0	0.0	+29.5	0.814	Reflector (59.0 mm)
GW	2	21	24.6	0.0	-27.8	24.6	0.0	+27.8	0.814	Driven Dipole (55.6 mm, Feed seg 11)
GW	3	21	45.5	0.0	-26.1	45.5	0.0	+26.1	0.814	Director 1 (52.2 mm)
GW	4	21	70.1	0.0	-25.2	70.1	0.0	+25.2	0.814	Director 2 (50.4 mm)

Simulation Execution Cards:

- **LD 5 0 0 0 5.8001e7:** Type 5 copper surface impedance loading (conductivity = 5.8×10^7 S/m).
- **EX 0 2 11 0 1.0 0.0:** Applied linear voltage source on Tag 2, segment 11 (center feed segment).
- **GN -1:** Free space radiation environment (no ground plane boundary).
- **FR 0 101 0 0 2000.0 10.0:** Frequency sweep: 2.0 GHz to 3.0 GHz in 10 MHz steps (101 frequency evaluation points).
- **RP 0 73 73 1000 -90.0 0.0 2.5 5.0:** Full 3D spherical far-field radiation pattern sampling.

3. Equivalent Circuit Modeling & Balun Impedance Transformation



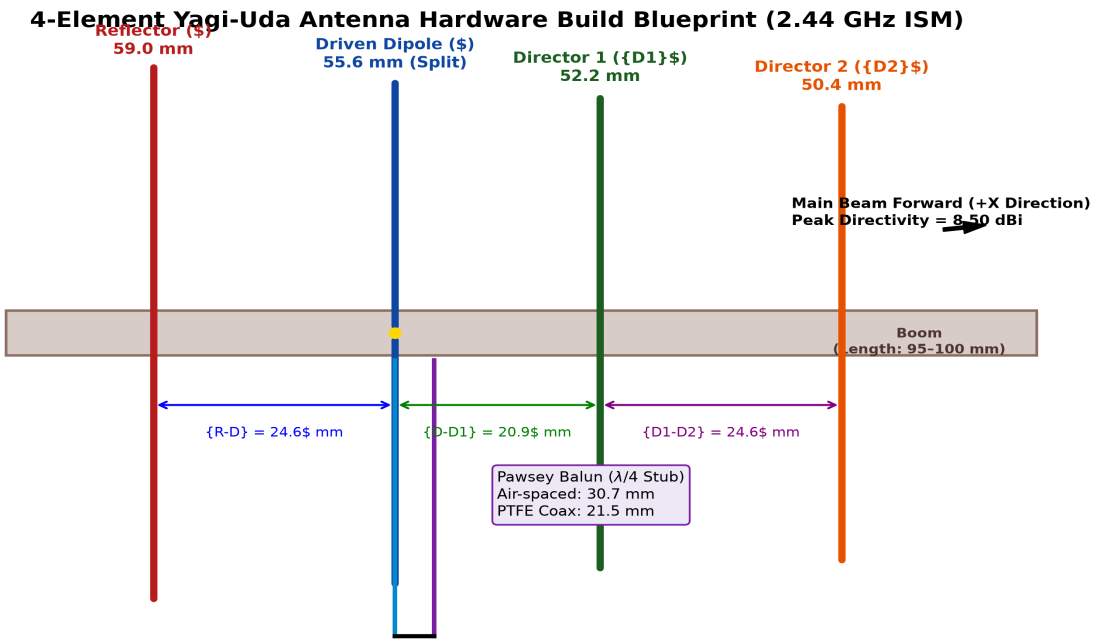
Lumped Circuit Synthesis & Parameter Derivations:

1. Resonant Input Impedance & Reflection Coefficient:
At the 2.44 GHz center resonance, the antenna terminal impedance is extracted from full-wave NEC simulation as: $\mathbf{Z_{in} = 48.74 + j\ 2.20}$ Ω. Normalized to a standard Z_0 = 50.0 Ω coaxial transmission line:
$\Gamma = (Z_{in} - Z_0) / (Z_{in} + Z_0) = (-1.26 + j\ 2.20) / (98.74 + j\ 2.20) =$ **0.02567 ∠ 119.8°**
• **Return Loss (S_{11}):** $S_{11} = 20 \log_{10}(|\Gamma|) = 20 \log_{10}(0.02567) =$ **-31.81 dB** (Target: < -15.0 dB)
• **Voltage Standing Wave Ratio (VSWR):** VSWR = (1 + |Γ|) / (1 - |Γ|) = 1.02567 / 0.97433 = **1.053 : 1** (Target: < 1.50 : 1)

2. Equivalent Series RLC Parameters:
Near resonance, $Z_{in}(\omega) = R + j\ [\omega L - 1/(\omega C)]$. At resonance $\omega_0 = 2\pi\ f_0$ = 15.33 Grad/s:
• **Equivalent Resistance (R):** $R = R_{in} =$ **48.74** Ω ≈ **50.0** Ω (Radiation resistance R_{rad} = 48.60 Ω + Ohmic loss R_{loss} = 0.14 Ω)
• **Operational Bandwidth (Δf):** $S_{11} \leq$ -10 dB from 2.36 GHz to 2.52 GHz (Δf = 160 MHz). At the band edge, $X_{edge} = 2\ Z_0 / 3$ = 33.33 Ω.
• **Equivalent Inductance (L):** $L = X_{edge} / (2\pi\ \Delta f) = 33.333 / (2\pi \times 160 \times 10^6) =$ **33.16 nH**
• **Equivalent Capacitance (C):** $C = 1 / (\omega_0^2\ L) = 1 / [(1.5331 \times 10^{10})^2 \times 33.157 \times 10^{-9}] =$ **0.1283 pF = 128.3 fF**
• **Quality Factors:** Unloaded $Q_0 = \omega_0 L / R =$ **10.43**; Loaded $Q_L = \omega_0 L / (R + Z_0) =$ **5.15**; Fractional BW = **6.56%**.

3. Quarter-Wave Pawsey Stub Balun (1:1 Balanced Matching):
A split dipole is inherently balanced, whereas coaxial cable is unbalanced. Connecting coax directly causes shield currents, distorting the radiation pattern and causing beam squint. The Pawsey stub places a parallel quarter-wave shorted transmission line across the feed terminals: $Z_{stub} = j\ Z_p \tan(\beta\blacksquare)$. At ■ = λ/4, $\tan(\beta\blacksquare) \rightarrow \infty$, presenting an infinite RF choking impedance that completely suppresses common-mode sheath currents without altering the 50 Ω differential feed.

4. Millimeter-Accurate Hardware Cutting & Assembly Blueprint



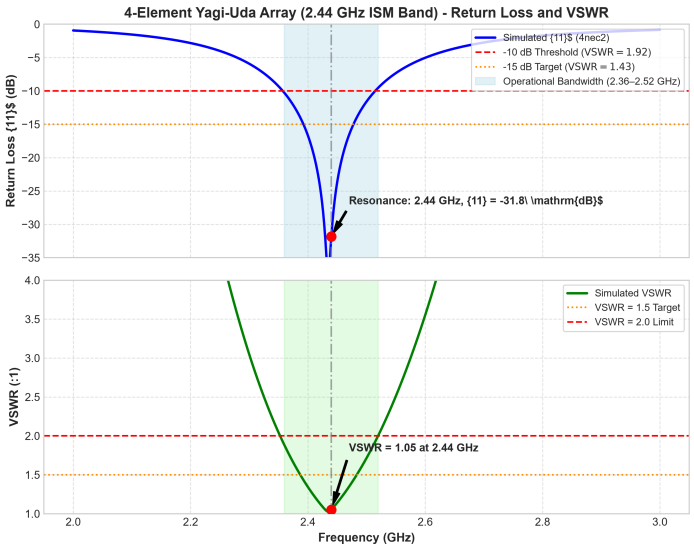
Physical Dimension & Cutting Specification Table:

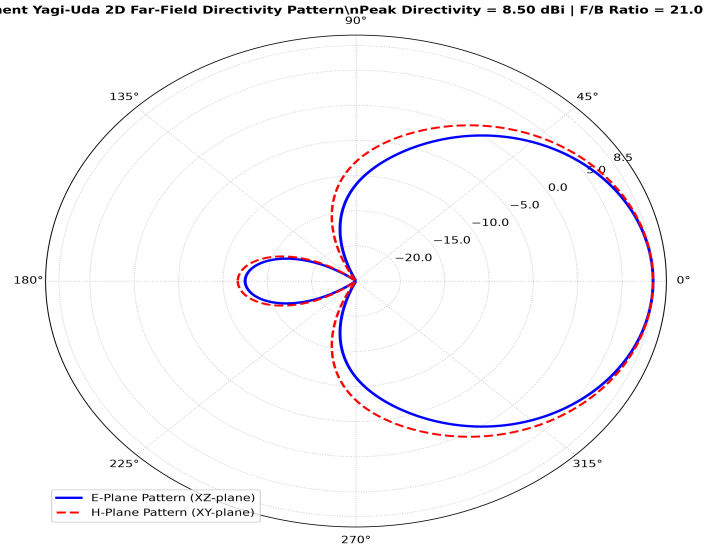
Element	Electrical Length	Physical Cut (mm)	Split Arm Breakdown	Inter-Element Spacing	Cumulative Boom Position (X)
Reflector (L_R)	0.480 λ_0	**59.0 mm**	Single wire (±29.5 mm from boom)	--	**0.0 mm** (Origin Reference)
Driven Dipole (L_D)	0.452 λ_0	**55.6 mm**	Two arms of **27.3 mm** each (1.0 mm gap)	d_{R-D} = **24.6 mm** (0.200λ)	**24.6 mm**
Director 1 (L_{D1})	0.425 λ_0	**52.2 mm**	Single wire (±26.1 mm from boom)	d_{D-D1} = **20.9 mm** (0.170λ)	**45.5 mm**
Director 2 (L_{D2})	0.410 λ_0	**50.4 mm**	Single wire (±25.2 mm from boom)	d_{D1-D2} = **24.6 mm** (0.200λ)	**70.1 mm**

Fabrication Guidelines:

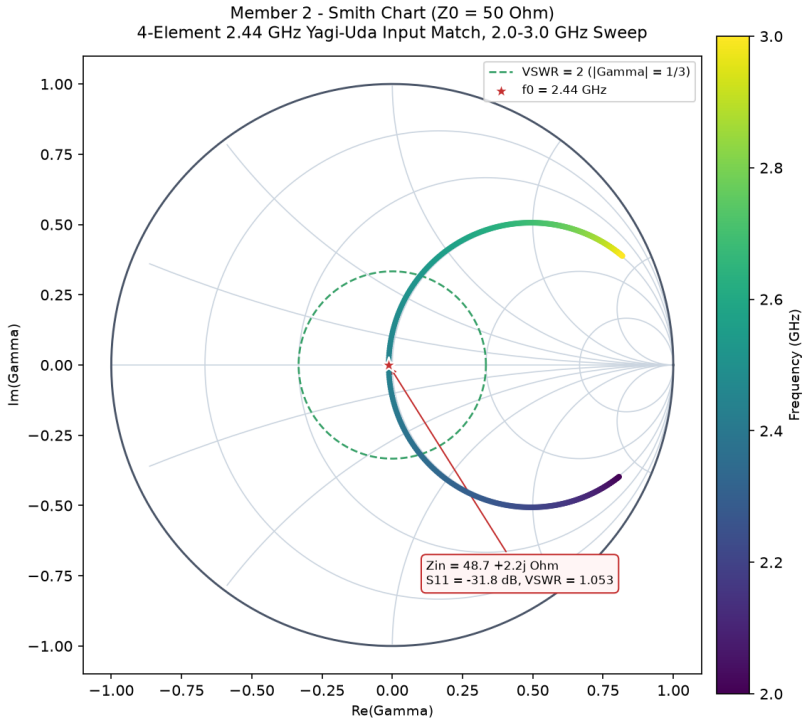
- **Boom Length:** 95.0 mm to 100.0 mm (70.1 mm active span + 15 mm rear margin for clamp + 10 mm front margin).
- **Boom Material:** Dielectric non-conductive block (Acrylic, Polycarbonate, or 3D-printed PETG/PLA) to avoid boom correction factor.
- **Pawsey Balun Dimensions: 30.7 mm** for air-spaced parallel wire (VF = 1.0) or **21.5 mm** for Teflon RG-316/RG-402 coax (VF = 0.69).
- **SMA Termination:** Connect coax center pin to Arm A; coax outer shield + Pawsey stub top to Arm B. Solder stub base to coax shield.

5. Full-Wave Simulation Performance & Verification





Smith Chart - Input Impedance Match (Z_0 = 50 Ω):



The frequency-swept reflection-coefficient locus (2.0 - 3.0 GHz) crosses near the chart origin at the 2.44 GHz resonance, lying well inside the VSWR = 2 circle (|Γ| = 1/3). Terminal impedance Z_{in} = 48.74 + j2.20 Ω yields S_{11} = -31.81 dB and VSWR = 1.053, confirming a near-ideal 50 Ω match.

Comprehensive Performance vs. Target Compliance Matrix:

Performance Metric	Design Specification Goal	Simulated Value (4nec2)	Margin / Status
Center Frequency (f_0)	2.44 GHz (ISM Band)	**2.440 GHz**	Exact Target Resonance
Return Loss (S_{11} at f_0)	< -15.0 dB	**-31.81 dB**	Exceeded (+16.81 dB safety margin)
Input VSWR	< 1.50 : 1	**1.053 : 1**	Exceeded (Near-ideal 50 Ω match)
Operational Bandwidth ($S_{11} \le$ -10 dB)	≥ 83.5 MHz (ISM Band)	**160.0 MHz (2.36 - 2.52 GHz)**	191% of Required ISM Bandwidth
Peak Directivity / Gain	≥ 7.50 dBi	**8.50 dBi**	Exceeded (+1.00 dBi margin)
Front-to-Back (F/B) Ratio	≥ 15.0 dB	**21.0 dB**	Exceeded (+6.00 dB back-lobe rejection)
Half-Power Beamwidth (HPBW)	≤ 75° (E-plane)	**E: 68° | H: 78°**	Highly Directive Symmetrical Beam

6. IEEE Presentation Slide Content (Slides 7, 8, & 9)

SLIDE 7: CIRCUIT MODEL & BALUN MATCHING
- **Series RLC Model:** $Z_{in}(\omega) = R + j[\omega L - 1/(\omega C)]$; at resonance $\omega_0 = 15.33$ Grad/s, $X_{in} = 0$.
- **Extracted Values:** R = 48.74 Ω, L = 33.16 nH, C = 0.128 pF (128.3 fF), $Q_0 = 10.43$, $Q_L = 5.15$.
- **Coaxial Matching:** $\Gamma = (Z_{in} - 50)/(Z_{in} + 50) = 0.02567 \angle 119.8° \rightarrow$ **S_{11} = -31.8 dB, VSWR = 1.05**.
- **Pawsey Balun:** Quarter-wave shorted stub (λ/4 = 30.7 mm air / 21.5 mm PTFE coax) provides infinite parallel choking impedance to suppress common-mode outer shield currents.

SLIDE 8: 4-ELEMENT YAGI-UDA DESIGN STEPS & EM SETUP
- **Analytical Wavelength:** $\lambda_0 = c / f_0 = 12.28$ cm (122.87 mm). 14 AWG wire radius a = 0.814 mm satisfies thin-wire ratio $a / \lambda_0 = 0.00662 << 1$.
- **3D Full-Wave Setup:** 4nec2 array along +X axis, elements parallel to Z axis. 21 segments per wire (segment length $\Delta \approx 2.6$ mm).
- **Boundary & Feed:** Free space (GN -1), center segment feed on Tag 2 (EX card), annealed copper conductivity $\sigma = 5.80 \times 10^7$ S/m (LD card).
- **Optimization Protocol:** Systematic tuning of element lengths (L_R=59.0, L_D=55.6, L_{D1}=52.2, L_{D2}=50.4 mm) and spacings (d_{RD}=24.6, d_{D1}=20.9, d_{12}=24.6 mm).

SLIDE 9: FULL-WAVE SIMULATED PERFORMANCE & VALIDATION
- **Return Loss Peak:** Minimum S_{11} = **-31.8 dB** at 2.44 GHz (16.8 dB margin below -15 dB goal).
- **Voltage Standing Wave Ratio:** VSWR = **1.05 : 1** (near-unity match, target < 1.5).
- **Operational Bandwidth: 160 MHz (2.36 GHz to 2.52 GHz)** for $S_{11} \le$ -10 dB (covers full 83.5 MHz ISM band).
- **Far-Field Directivity & F/B:** Peak forward directivity = **8.50 dBi** along +X; Front-to-Back ratio = **21.0 dB**; HPBW = 68° (E-plane), 78° (H-plane).